SR series
Industrial Robot

TRANSPORTATION AND INSTALLATION MANUAL

(SEPARATE VOLUME)

(SR–554HSP–T–Z–S)

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
   After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

NUMAZU, JAPAN

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PREFACE

This manual constitutes a separate volume of the SR–554HSP industrial robot transportation and installation manual.
It describes the differences between the SR–554HSP–T specifications and the standard specifications.
For the general descriptions, you can refer to the Transportation and Installation Manual given below, in addition to this manual.
Be sure to look through this manual before actually unpacking the shipment.

- Transportation and Installation Manual (STE 71372)

PRECAUTIONS FOR SAFETY

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 Danger	This means that "incorrect handling will lead to fatalities or major injuries".
 Caution	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited(must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory(must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. The details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

Caution

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

Danger	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
 Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
 Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause mis-operation by noise.

Caution	
 Prohibited	• NEVER lift the robot by the arm 2 cover or arm 2. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	• When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, securely fix it on the base. The robot will be unstable if just set down, and it could tilt over.

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Section 1 Transportation

1.1 Transporting the Robot

In principle, the robot is shipped with the arm folded back and secured with the attached clamp as in the standard robot. (After you have unpacked the shipment, you can move it as it is.)

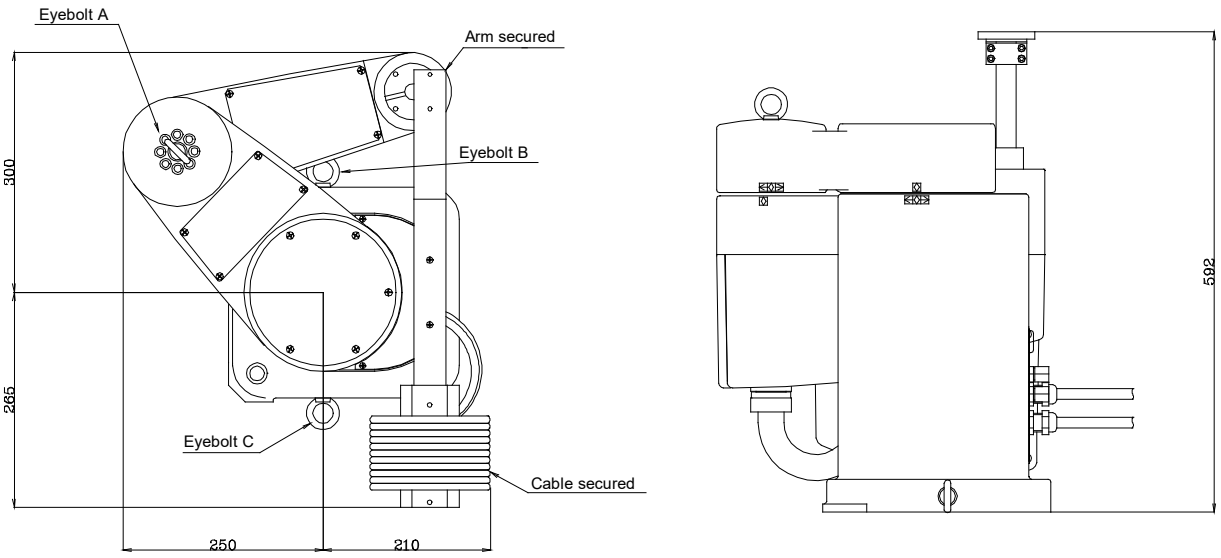


Fig. 1.1 Posture when the robot is to be transported

It is possible to lift up and carry the robot. Pass wires through eyebolts A, B and C shown in the figure above, then lift up the robot carefully.

Mass of the robot: Approx. 55 kg

Section 2 Installation

2.1 Robot Installation

Before installing the robot, you should plan a layout, fully considering the robot working envelope, coordinate system and space for maintenance of the robot.

2.1.1 External Dimensions

Refer to the separate drawing (Dwg. No. S–26F04).

2.1.2 Working Envelope

Fig. 2.1 shows the working envelope of the robot.

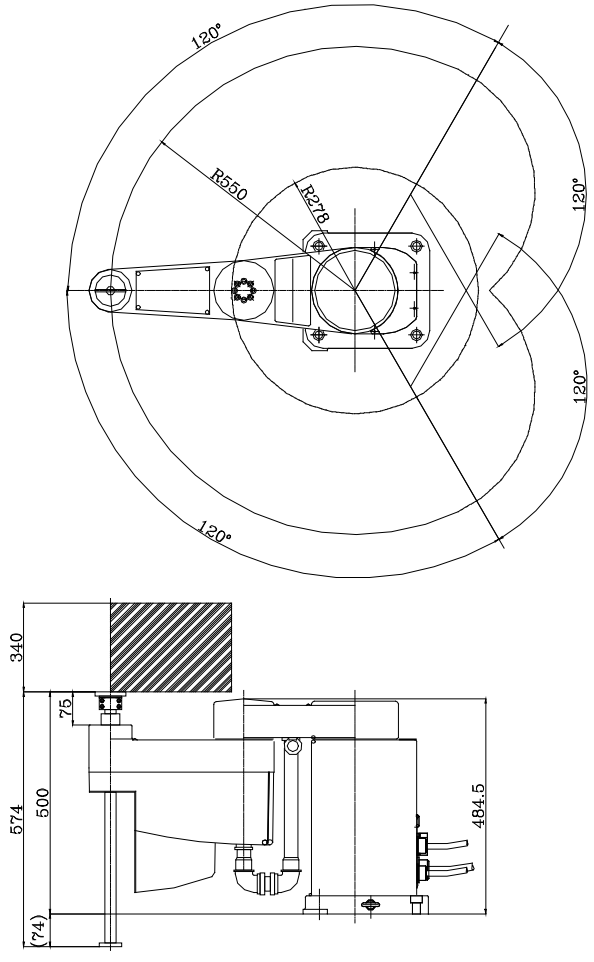


Fig. 2.1 Working envelope of the robot

2.1.3 Coordinate System

The robot's joint angle zero point (0° or 0 mm position) is factory-calibrated according to its base reference planes. Fig. 2.2 shows the base coordinate system and zero position of each axis joint angle.

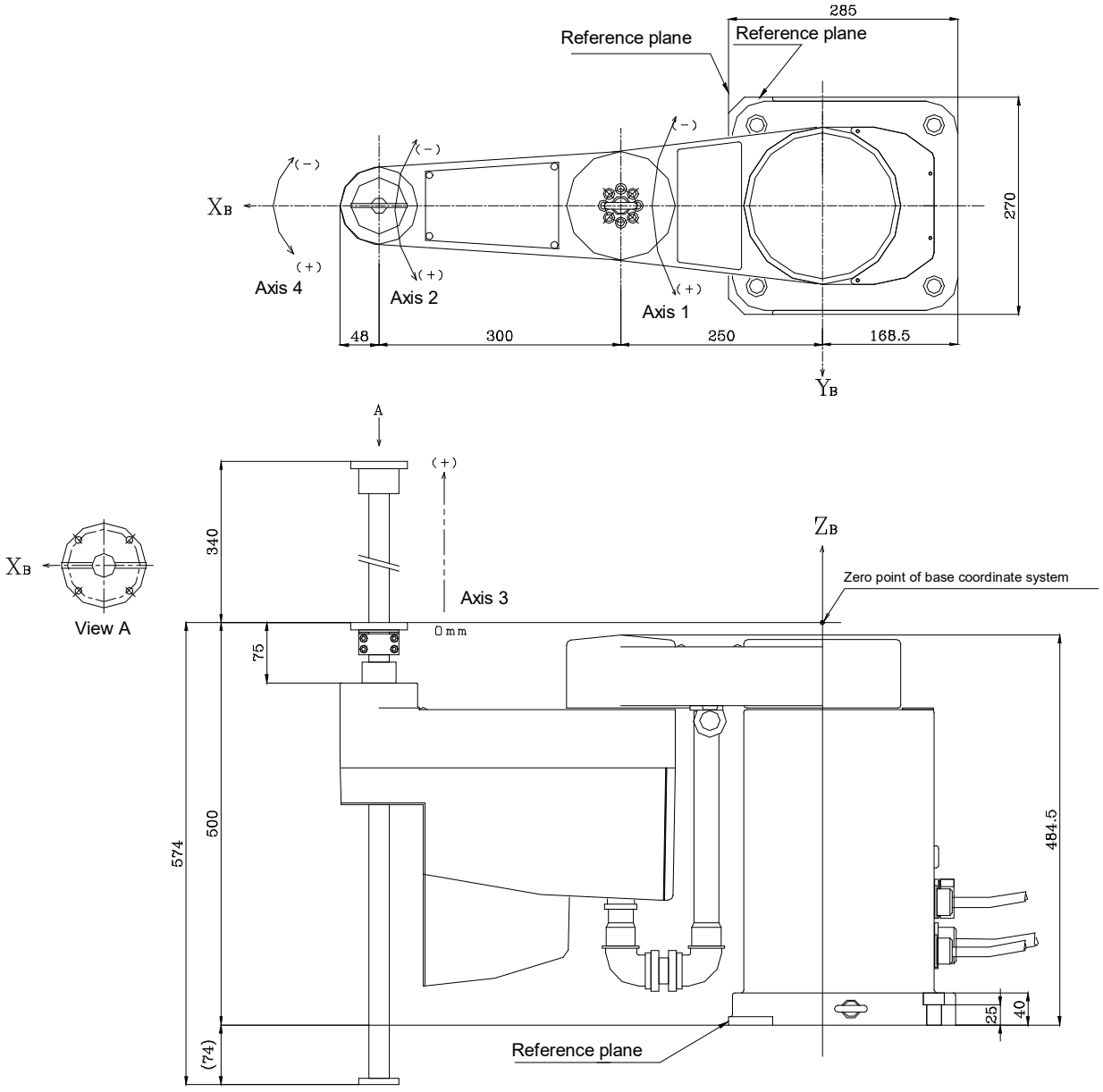


Fig. 2.2 Coordinate system

2.1.4 Installing the Robot

The robot is secured, using the set holes on the base. Use M16 hexagon socket head cap screws for the SR–554HSP–T–Z–S robot.
The robot installation method is shown in Fig. 2.3. Reference planes are provided on the base section and marked with "xxx".
To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot.
The reference planes are factory-coated with anticorrosive agent. Before use, wipe out the agent with thinner.

Caution

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, the vibration will occur while the robot is operating, and could lead to faults.

  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

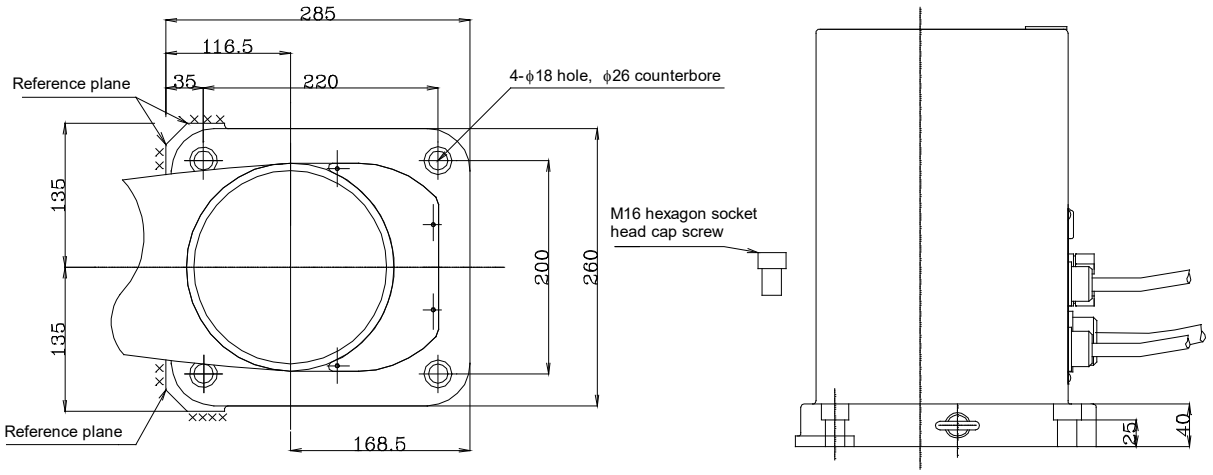


Fig. 2.3 Installing the robot